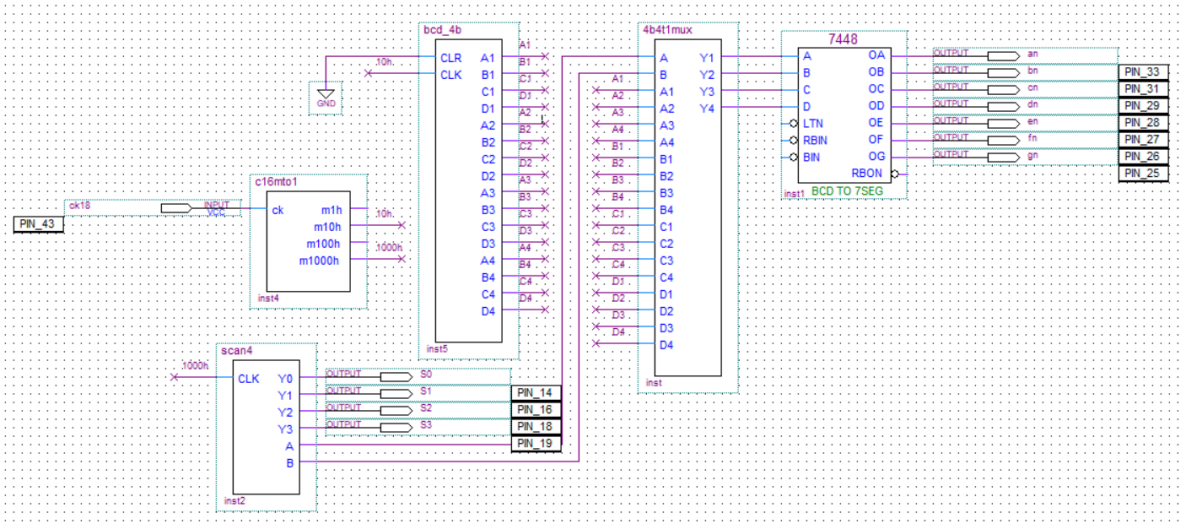
CPLD 可程式數位邏輯設計實習單元_數位式電子鐘

實習目的：

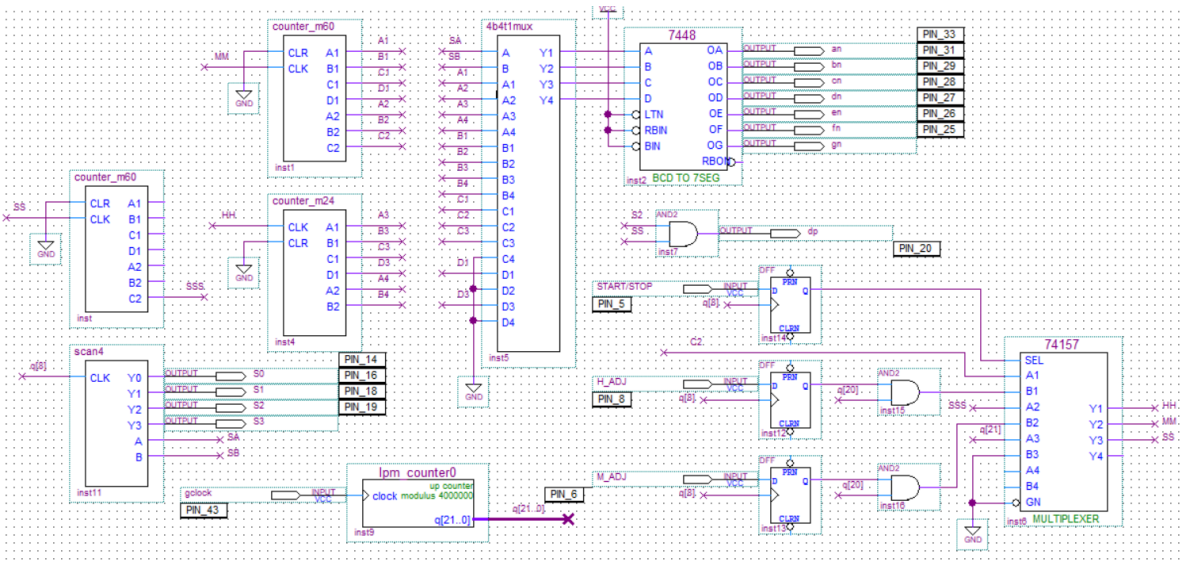
- 使學生由數位邏輯學習中，啟發應用組合邏輯的應用。
- 使學生善用解碼器、多工器、解多工器配上計數器(除頻器)組合。
- 使學生會應用 EDA 軟體及 KiCAD 繪圖軟體完成電路。

實習過程回顧：

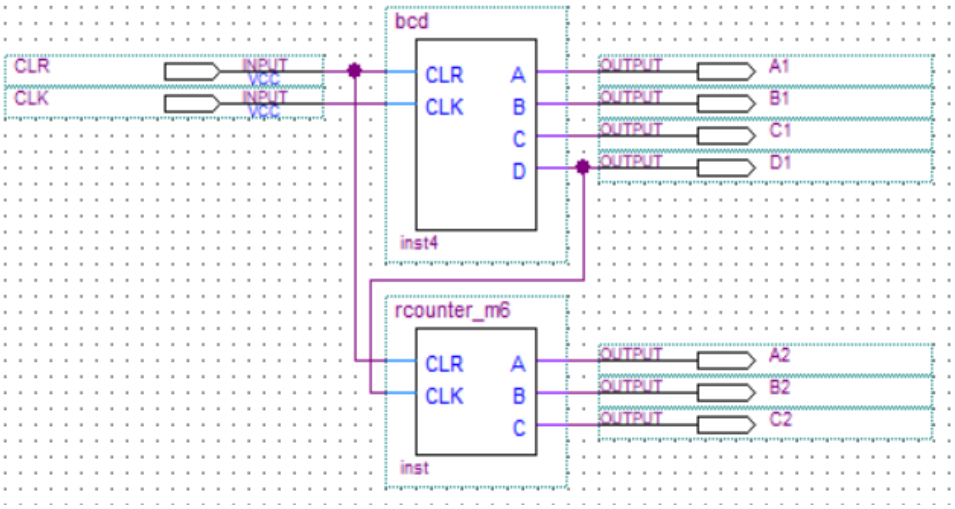
- 組合邏輯實驗
  - 基本邏輯閘認識與組合
- 多工器、解多工器、解碼器實驗
  - 2to1、4to1、8to1、2to4、3to8、4to16、七線段解碼器實驗
- 正反器實驗
  - RS-FF、JK-FF、D-FF、T-FF 實驗
- 計數器實驗
  - 同步計數器、非同步計數器(漣波計數器)實驗
    - mod3、mod6、mod10、mod60、mod24 計數電路實驗
    - 7490、74490、74294、LPM 計數模塊應用
- 四位多工掃描顯示電路組合



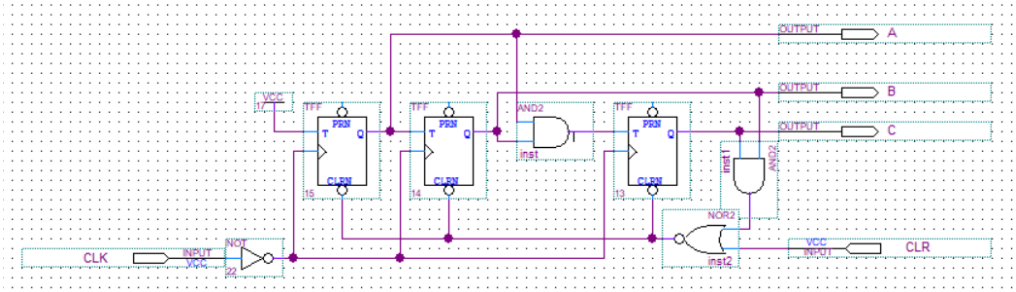
- 應用計數模塊組合_數位式電子鐘



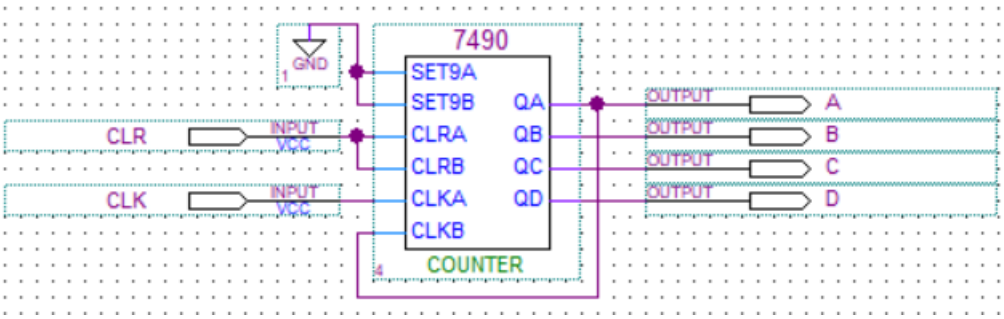
■ **counter m60**



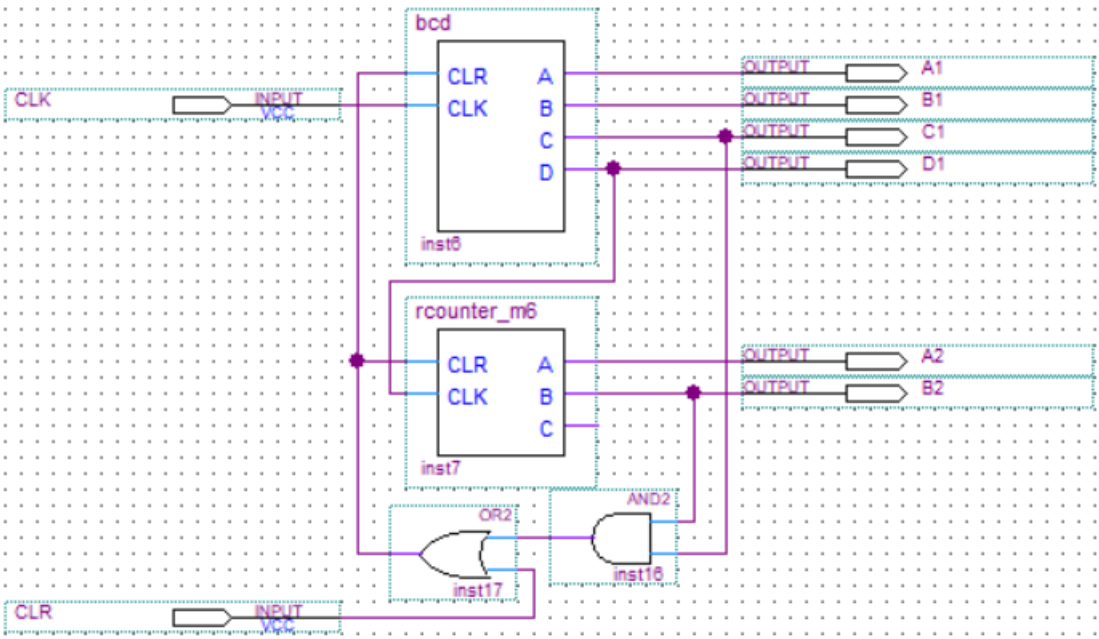
◆ **rcounter m6**



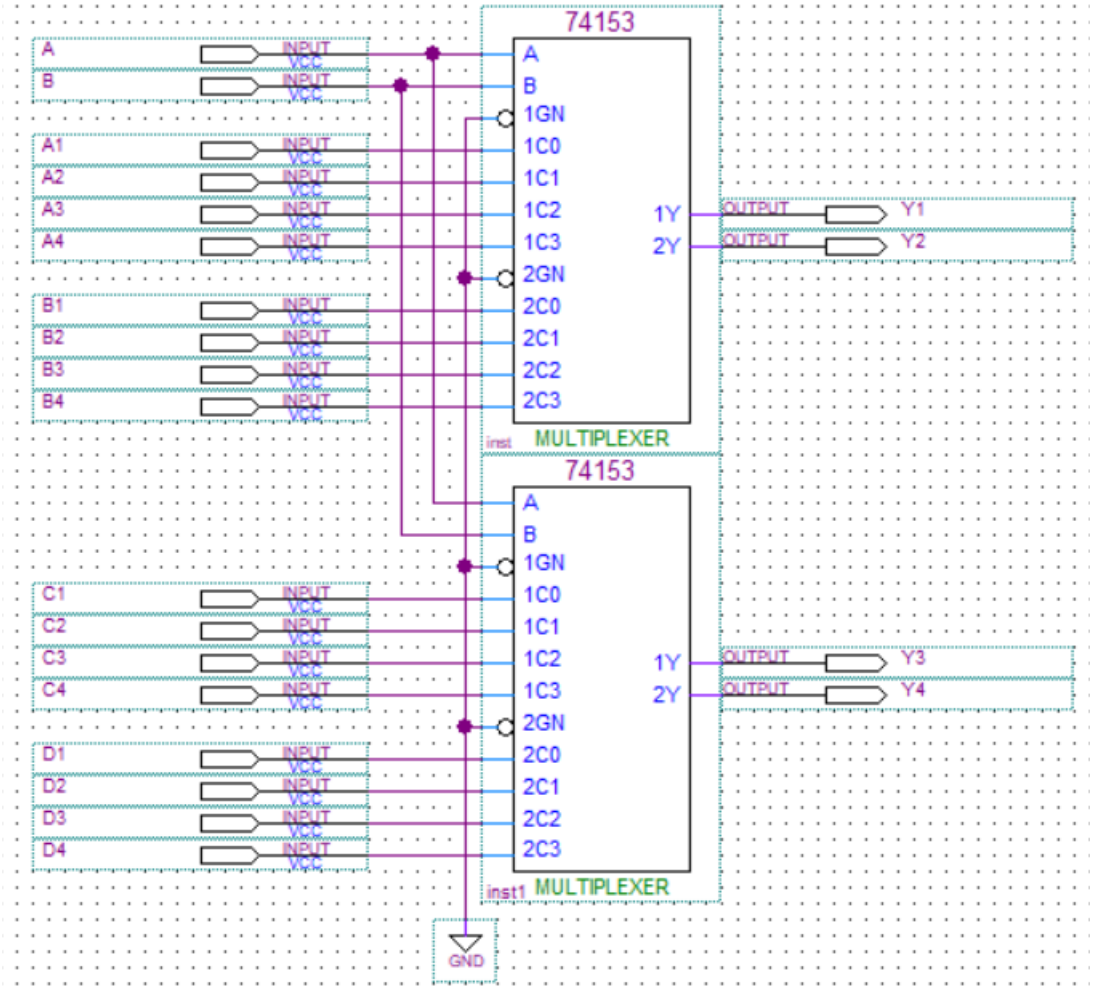
◆ **bcd**



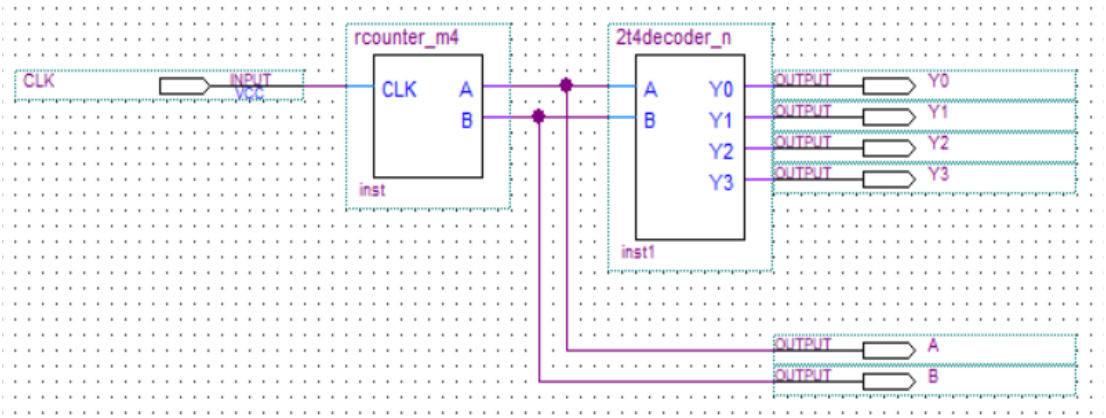
■ **counter m24**



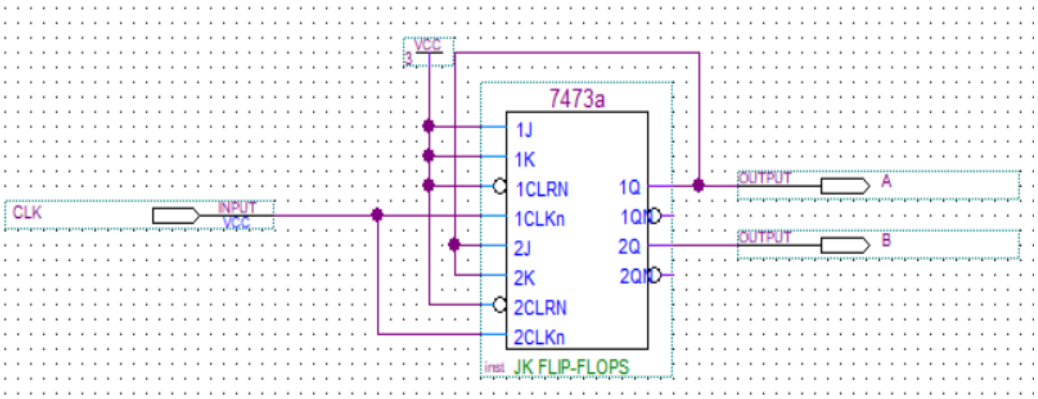
■ **4b4t1mux**



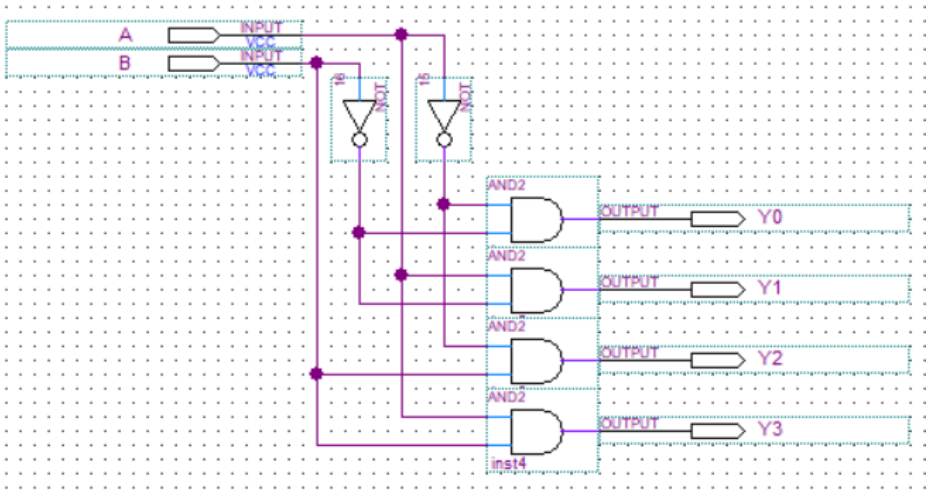
■ **scan4**



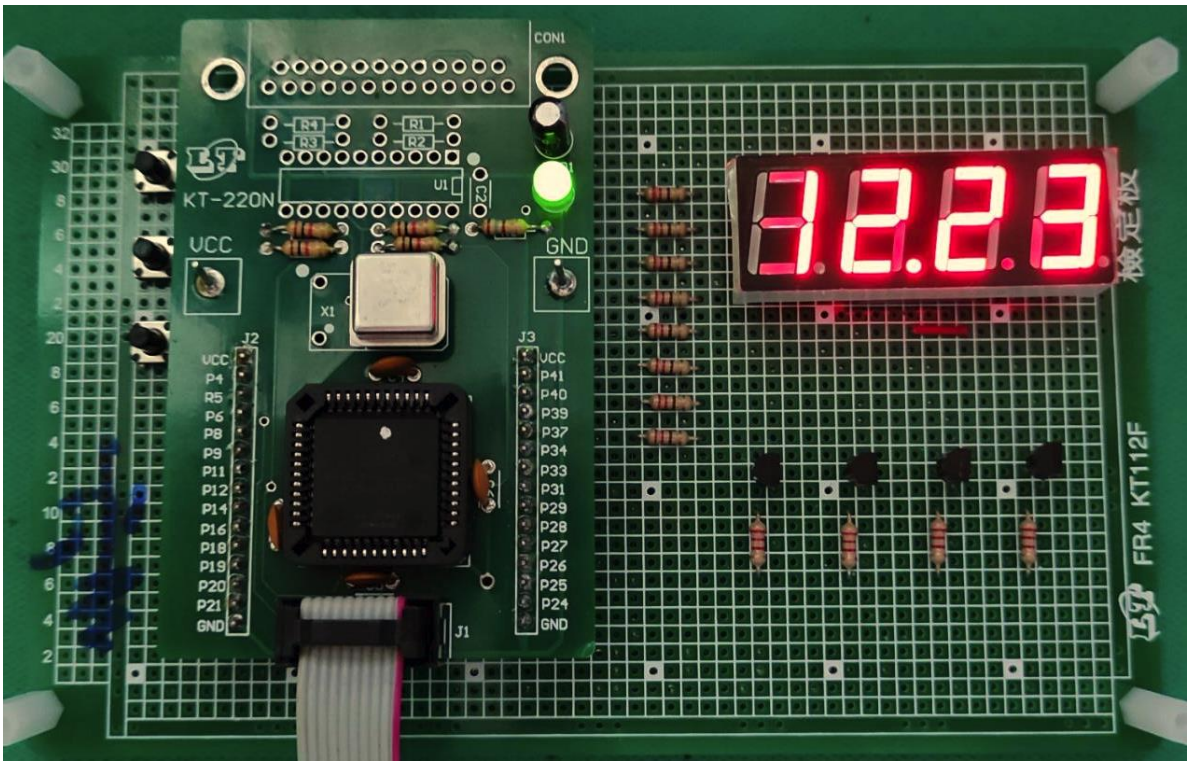
◆ **rcounter m4**



◆ **2t4decoder_n**



■ **7448 IC 共陰極七線段解碼器**

■ **74157 IC 四組二對一多工器**

● **實體電路**